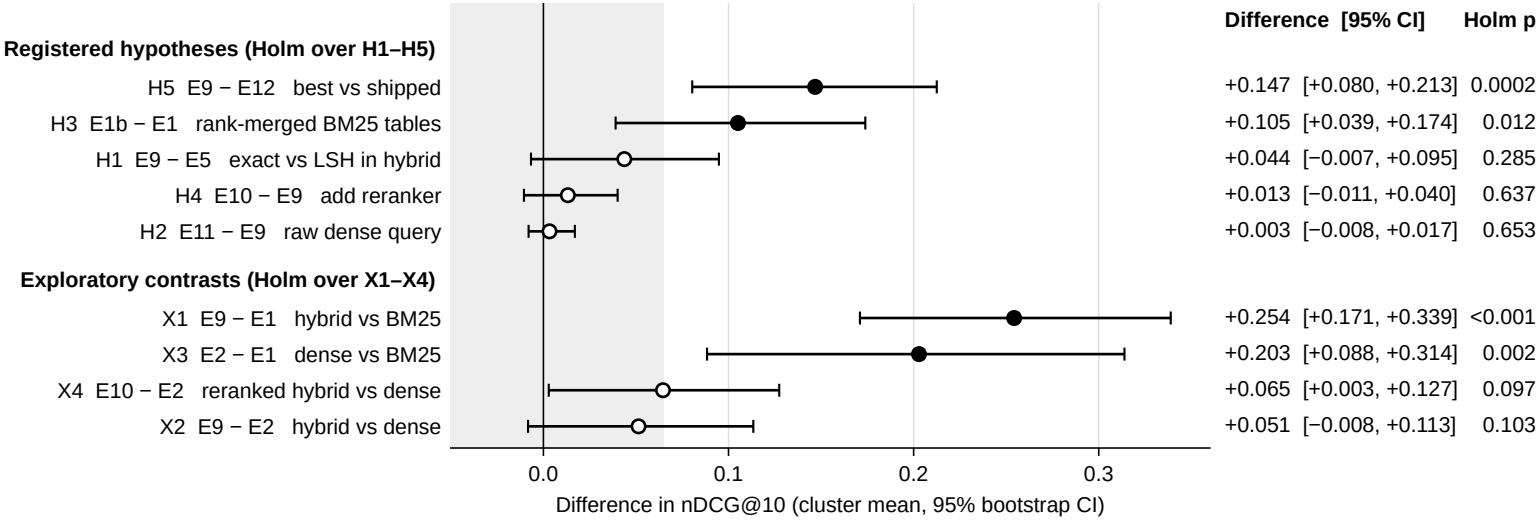
Registered hypotheses (Holm over H1–H5)
H5 E9 − E12 best vs shipped
H3 E1b − E1 rank-merged BM25 tables
H1 E9 − E5 exact vs LSH in hybrid
H4 E10 − E9 add reranker
H2 E11 − E9 raw dense query
Exploratory contrasts (Holm over X1–X4)
X1 E9 − E1 hybrid vs BM25
X3 E2 − E1 dense vs BM25
X4 E10 − E2 reranked hybrid vs dense
X2 E9 − E2 hybrid vs dense
Difference [95% CI] Holm p
+0.147 [+0.080, +0.213] 0.0002
+0.105 [+0.039, +0.174] 0.012
+0.044 [−0.007, +0.095] 0.285
+0.013 [−0.011, +0.040] 0.637
+0.003 [−0.008, +0.017] 0.653
+0.254 [+0.171, +0.339] <0.001
+0.203 [+0.088, +0.314] 0.002
+0.065 [+0.003, +0.127] 0.097
+0.051 [−0.008, +0.113] 0.103
0.0
0.1
0.2
0.3
Difference in nDCG@10 (cluster mean, 95% bootstrap CI)